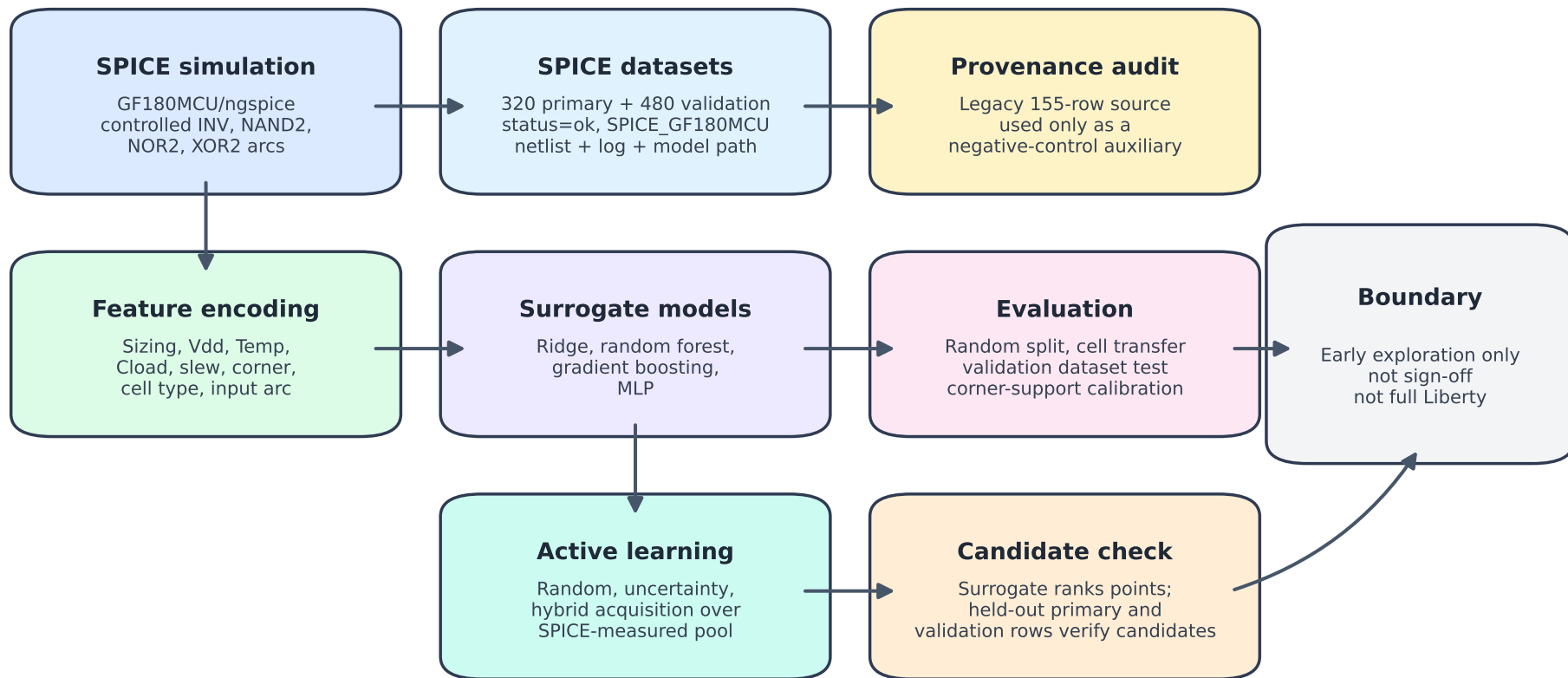


Source-aware SPICE-efficient surrogate workflow

Early design-space exploration with explicit provenance and SPICE re-validation



Final design points remain subject to conventional SPICE re-simulation under the intended PDK, PVT, load, slew, and layout conditions.